

Algebraic Systems.

Semigroups - Monoids - Groups

Let G be a non-empty set

Let $*$ be a binary operation on G

(a map $*$: $G \times G \rightarrow G$)
or function

Condition 1: Associativity.

for all x, y and z in G

$$x * (y * z) = (x * y) * z$$

Condition 2: Identity

there exists an element $e \in G$ such that

for every $x \in G$

$$x * e = e * x = x$$

Condition 3: Inverse

for every $x \in G$, there exists an element

(denoted) $x^{-1} \in G$ such that

$$x * x^{-1} = x^{-1} * x = e$$

Semigroup, $(G, *)$ said to be semigroup if it satisfies Condition 1

$(G, *)$ is said to be Monoid if it satisfies

Conditions 1 & 2

$(G, *)$ is said to be a Group if it satisfies

Conditions 1, 2 and 3.

$(\mathbb{E}, +)$, $\mathbb{E}$ - set of ⁺even numbers.

$(\mathbb{E}, \times)$ are semigroups. but not Monoids.

$(\mathbb{N}, +)$ $\mathbb{N}$ - natural numbers.

→ semigrp, not monoid

$(\mathbb{N}, \times)$ → semigrp, monoid. but not group

$(\mathbb{I}, +)$, $\mathbb{I}$ set of integers

→ semigrp, monoid and group

Set of rational numbers excluding zero
w.r.t multiplication.

→ semigrp, monoid and group.

Group Properties.

Let $(G, *)$ be a group.

1. In a group G , the identity elt is unique

Let e_1 and e_2 be two identity elts in G .

$$\text{Now } \xrightarrow{(e_2 \text{ is identity})} e_1 = e_1 * e_2 = e_2 \xrightarrow{(e_1 \text{ is identity})}$$

$\therefore$ group G has unique identity elt

2. In a group $(G, *)$, Inverse of any element is unique.

Let $a \in G$ and a_1 and a_2 are two inverses of a .

$$\begin{aligned} i) \quad a * a_1 &= a_1 * a = e \xrightarrow{\text{identity}} \\ a * a_2 &= a_2 * a = e \end{aligned}$$

$$\begin{aligned} \text{Now } a_1 &= a_1 * e \\ &= a_1 * (a * a_2) \\ &= (a_1 * a) * a_2 \quad \text{by associativity.} \\ &= e * a_2 \\ &= a_2 \end{aligned}$$

3. Cancellation laws

$$\begin{aligned} i) \quad a * b &= a * c \implies b = c \\ ii) \quad b * a &= c * a \implies b = c \end{aligned}$$

Pf: i) Let

$$\begin{aligned} a * b &= a * c \\ a^{-1} * (a * b) &= a^{-1} * (a * c) \quad a^{-1} \text{ is the inverse of } a \\ (a^{-1} * a) * b &= (a^{-1} * a) * c \quad \text{associativity} \\ e * b &= e * c \quad (e \text{ identity}) \\ b &= c \end{aligned}$$

Standard Notations:

$(G, *)$ — group.

$a \in G$

Define.

$$a^0 = e, \quad a^{n+1} = a^n * a \quad \text{for } n \in \mathbb{N}$$

$$(a^{-1})^n = a^{-n} \quad \text{for } n \in \mathbb{N} \quad \text{natural nos}$$

$$\text{also } a^m * a^k = a^{m+k} \quad \text{for } m, k \in \mathbb{I} \quad \text{Integers}$$